

Modeling the Universal Set/Reset Characteristics of Bipolar RRAM by Field- and Temperature-Driven Filament Growth

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Abstract—Resistive switching memory (RRAM) devices generally rely on the formation/dissolution of conductive filaments through insulating materials, such as metal oxides and chalcogenide glasses. Understanding the mechanisms for filament formation and disruption in resistive switching materials is a critical step toward the development of reliable and controllable RRAM for future-generation storage. In particular, the capability to control the filament resistance and the reset current through the compliance current during filament formation may provide a key signature to clarify the switching mechanism. This paper provides a physically based explanation for the universal resistance switching in bipolar RRAM devices. A numerical model of filament growth based on thermally activated ion migration accounts for the resistance switching characteristics. The same physical picture is extended to numerically model the reset transition. The impact of migration parameters and experimental setup on the set/reset characteristics is discussed through numerical simulations.

Index Terms—Device modeling, metal–insulator transition, nonvolatile memory, oxide electronics, resistive switching memory (RRAM).

I. INTRODUCTION

RESISTIVE SWITCHING MEMORY (RRAM) devices are receiving widespread interest for future nonvolatile and dynamic memory technologies [1]. For the development of a practical and reliable RRAM technology, however, various severe challenges need to be addressed and solved, including switching variability [2], reset current reduction [3]–[6], and reliability [7]–[9]. Toward this goal, a sound physical understanding and modeling of the resistive switching mechanism is clearly mandatory.

Except for some ternary oxides showing a change in resistance due to interface switching [10], [11], most binary metal oxides and chalcogenide glasses evidence filament formation/dissolution as the key mechanism for resistance switching [1], [12]–[15]. The formation and disruption of a conductive filament (CF) result in transitions to low and high

resistances, respectively, which are referred to as set and reset processes. Filamentary switching has been demonstrated under both unipolar (both set and reset under positive voltage [3], [4], [12], [15]) and bipolar (positive/negative voltages for set/reset processes, or vice versa [2], [8]) modes, sometimes coexisting within the same material [16]–[18]. From the modeling viewpoint, a key feature of filamentary switching is the controllability of set and reset characteristics through the current compliance, namely, the maximum current I_C supplied to the device during the set transition [2]–[4], [19]–[24]. A universal set characteristic was demonstrated, where all the devices show a very similar behavior of the set-state resistance R after set following the inverse proportionality $R \propto I_C^{-1}$, irrespective of material composition, experimental setup, and applied (set) voltage [25]. In turn, the reset current I_{reset} needed to dissolve the CF is inversely proportional to R , hence proportional to I_C [3], [4], [25]. The inverse proportionality between I_{reset} and R can be understood by a law of constant reset voltage V_{reset} , where the value of V_{reset} is dictated by the critical temperature for the onset of filament dissolution by oxidation and/or ion migration/diffusion [12], [25]. The universal reset characteristic, namely, the weak material dependence of V_{reset} , was explained by the weak dependence of the reset temperature (hence voltage) on ion migration parameters, namely, diffusivity, mobility, and activation energy [25]. There is wide agreement on the critical role of the migration of ions, namely, oxygen vacancies [26]–[28] or cations [1], [24], [28], [29]. However, the detailed mechanism for filament formation and the strong dependence of R on I_C have not been explained to date.

This paper provides a physical model for filament growth in RRAM devices showing bipolar switching, allowing to explain the dependence of set-state resistance and reset current on the compliance current during set. A thermally activated hopping model is used to describe the ion migration and the consequent filament growth during set transition. The model allows highlighting the self-limiting growth of the filament that occurs in the presence of a load resistance or transistor in series with the RRAM device or under the effect of a current compliance system. The same model for thermally activated ion migration is extended to describe the reset process. Numerical calculations are shown to account for experimental data from the literature and allow exploring the dependence of set/reset characteristics on kinetic/material parameters and experimental measurement conditions.

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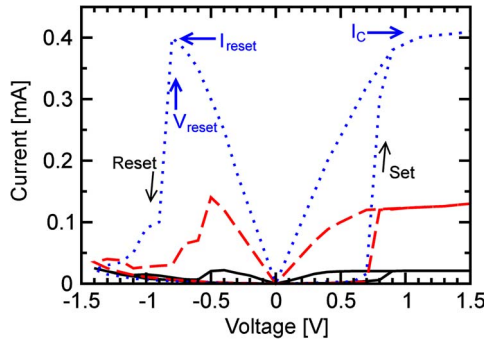


Fig. 1. (Color online) Measured I - V characteristics for set (positive voltage) and reset (negative voltage) for an HfO_x MIM [19]. The cell was connected to a MOS transistor in the so-called 1T1R structure to limit the current during set. The absolute value of the current is shown. Different colours correspond to different I_C , that is the maximum current during set. The set state resistance decreases and the reset current increases for increasing I_C .

II. SET-RESET CHARACTERISTICS

Fig. 1 shows the current-voltage (I - V) characteristics for bipolar RRAM devices based on metal-insulator-metal (MIM) structures, with HfO_x ($x = 1.4$) as insulating material [19]. The application of a positive voltage above the set voltage V_{set} to the high resistance state results in the set transition, consisting of the formation or regrowth of a low resistivity CF, similar to the dielectric breakdown effect [12]. The maximum current during set was maintained equal or below a compliance current I_C by the series connection of the RRAM device with a MOS transistor, where the saturated drain current was dictated by the applied gate voltage [19]. Such a one-transistor—one-resistor (1T1R) configuration allows suppressing uncontrollable current overshoots due to the parasitic capacitance of the RRAM device [3], [4]. Three set processes with different values of compliance current I_C are shown in the figure. The RRAM resistance R , determined from the I - V curve along the descending voltage sweep after set, decreases for increasing I_C . Application of a negative voltage to the low resistance state results in the reset operation at a characteristics reset current I_{reset} . The reset current increases with I_C (hence decreases with R), following $I_{\text{reset}} \approx I_C$. Similar behaviors were shown for other RRAM devices based on metal oxides ($\text{HfO}_x/\text{ZrO}_x$ [2], NiO [3], [4], [20], TiO_x [21]) or based on cation migration in Ag-doped GeSe [22]–[24] and Cu-doped SiO_2 [23].

III. ION MIGRATION MODEL

The relationship among I_C , R , and I_{reset} for bipolar switching RRAM devices in Fig. 1 can be explained by the model described in Fig. 2. The figure shows a MIM structure, where the insulator is assumed to consist of HfO_x , although any other switchable metal oxide may be assumed. The electrodes are assumed to be relatively inert metals, such as Pt, W, or TiN. Application of a positive voltage on the top electrode (TE) results in a field-driven migration of ions through the insulating layer. The ions depicted in the figure might consist of positively charged oxygen vacancies [26]–[28], or metallic cations, such as positively charged Hf ions with enhanced mobility due to the large local concentration of defects (vacancies, grain boundaries) [1], [24], [28], [29]. Note that this picture also includes

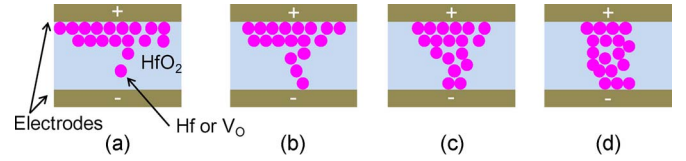


Fig. 2. (Color online) Sketch for the set model based on thermally activated ion migration and filament growth. Set takes place through (a) and (b) filament nucleation and (c) and (d) growth due to migration of positively charge ions from the TE, where a positive voltage is applied. Ions might consist of oxygen vacancies or metallic cations, namely, Hf ions in the case of HfO_x . The model also describes reset by the opposite sequence, namely, (c) and (d) CF narrowing and (a) and (b) final disconnection under a negative voltage applied to the TE. The reservoir of ions at the TE side might result from the previous reset operation.

the migration of negatively charged oxygen ions, which move in the direction opposite to their respective vacancies. Ion migration from the top to the bottom electrode results in the formation of a CF, according to two steps, namely, nucleation [Fig. 2(a) and (b)] and growth [Fig. 2(c) and (d)] [29]. The nucleation site might be dictated either by a preferential localization of the electric field due to a local thinning of the insulating layer or by a localization of the current due to threshold switching phenomena possibly enhanced by a locally higher concentration of defects [30], or by both. Note that a reservoir of ions is needed on the positive electrode side, which might consist of an accumulation of ions from the previous reset process. In the case of RRAM devices based on cation migration and solid-state electrolytes, the role of ion reservoir is played by one electrode of active metals (Ag, Cu), which can release ions for CF growth [1], [22]–[24], [28], [29]. After nucleation, the CF grows by ion migration and deposition (namely, reduction) at the CF surface [29]. Under opposite polarity (i.e., negative bias on the TE in Fig. 2), the reset transition follows the same sequence in Fig. 2 but in reverse order due to ion migration back to TE and consequent CF dissolution.

Neglecting the filament nucleation time, the kinetics of the set process can be described by the CF growth process driven by ion migration through thermally activated hopping. The CF growth rate is assumed proportional to the ion migration rate according to [31], [32]

$$\frac{d\phi}{dt} = A e^{-\frac{E_A}{kT}} \quad (1)$$

where ϕ is the CF diameter, assumed constant across the insulating layer, A is a preexponential constant, k is the Boltzmann constant, T is the effective CF temperature, and E_A is the activation energy, which is given by

$$E_A = E_{A0} - \alpha qV \quad (2)$$

where E_{A0} is the energy barrier for ion hopping, q is the elementary charge, V is the voltage across the insulating layer, and α is a barrier lowering coefficient [31]–[33]. The term αqV in (2) describes the barrier lowering due to the applied field, forcing the positive ions to move along the direction of the electric field. This is similar to the thermally activated hopping mechanism for electrons and holes in disordered semiconductors, where the exponential dependence of the current of voltage is explained by a Poole–Frenkel-like barrier lowering effect

[34]. The local temperature T in (1) is significantly impacted by Joule heating due to the considerable current density and voltage during set. To calculate the local temperature, we solve the steady-state Fourier equation given by

$$k_{th} \frac{d^2 T}{dz^2} + \frac{F^2}{\rho} = 0 \quad (3)$$

where k_{th} is the thermal conductivity, ρ is the electrical resistivity, F is the electric field, and z is the coordinate along the longitudinal axis of the CF. Transient effects in the heat conduction equation can safely be neglected due to the usually small value of the characteristic thermal time τ_{th} of the CF. The thermal time can be estimated as $\tau_{th} = R_{th} C_{th} \approx C_p L^2 / k_{th}$, where R_{th} is the effective thermal resistance of the CF [12], [25], [31], C_{th} is the thermal capacitance of the CF, L is the filament length, and C_p is the thermal capacitance of the CF material. A value $\tau_{th} = 33 \times 10^{-12}$ s is obtained assuming a filament of metallic Hf ($C_p = 1.92 \text{ JK}^{-1}\text{cm}^{-3}$, $k_{th} = 23 \text{ Wm}^{-1}\text{K}^{-1}$) with length $L = 20$ nm. Such a small τ_{th} is clearly irrelevant for practical set/reset transients. In addition, note that (3) only takes into account heat diffusion along the CF, thus neglecting the out-of-filament heat conduction through the insulating layer [34]. Solving (3) with boundary conditions $T = T_0$ (room temperature) at the two electrodes at distance L leads to a maximum temperature along the CF equal to [12], [34]

$$T = T_0 + R_{th} R I^2 = T_0 + \frac{R_{th}}{R} V^2 \quad (4)$$

where $I = V/R$ is the current through the filament. Using the formulas $R_{th} = L / (8k_{th} A_{CF})$ [12] and $R = \rho L / A_{CF}$, where $A_{CF} = \pi \phi^2 / 4$ is the CF cross-sectional area, yields

$$T = T_0 + \frac{2\rho L^2}{\pi^2 k_{th}} \frac{I^2}{\phi^4} = T_0 + \frac{V^2}{8\rho k_{th}} \quad (5)$$

which allows estimating the CF temperature based on the voltage across the cell or the current through the CF. Substitution of (2) and (5) into (1) leads to

$$\frac{d\phi}{dt} = A e^{-\frac{E_{A0} - \alpha q V}{k T_0 \left(1 + \frac{V^2}{8 T_0 \rho k_{th}}\right)}} \quad (6)$$

which can numerically be solved to simulate the CF growth during set. The reset process for CF dissolution can similarly be modeled by the same equations provided that a minus sign is inserted in the right-hand side of (1) and (6) to describe the CF narrowing (instead of growth) process [12].

IV. SIMULATION RESULTS

Calculations according to (6) were conducted for the set/reset processes in HfO_x -based RRAM [19], [35], assuming physical parameters $A = 1 \text{ ms}^{-1}$, $E_{A0} = 1.2 \text{ eV}$ [7], [12], $\alpha = 0.3$ [24], [31], $k_{th} = 20 \text{ Wm}^{-1}\text{K}^{-1}$, an as-nucleated initial filament diameter $\phi_N = 0.5 \text{ nm}$, and a CF resistivity $\rho = 400 \mu\Omega\text{cm}$. The initial CF was assumed to be only a few atoms wide after nucleation. The electrical resistivity value was chosen to be relatively large with respect to Hf bulk resistivity $\rho_{\text{bulk}} =$

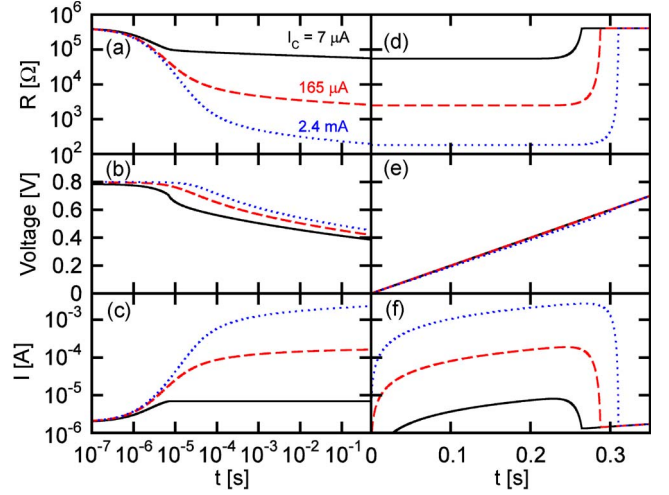


Fig. 3. (Color online) Calculated time evolution of (a) R , (b) V , and (c) I during set and the corresponding time evolution of (d) R , (e) V , and (f) I during the following reset transition for increasing gate bias of the MOS transistor, corresponding to three values of the current compliance $I_C = 7 \mu\text{A}$, $165 \mu\text{A}$, and 2.4 mA . A constant voltage $V_A = 0.8 \text{ V}$ was applied to the 1T1R sample during set, whereas a voltage sweep $V = -\beta t$ with $\beta = 2 \text{ Vs}^{-1}$ was applied for reset.

$33 \mu\Omega\text{cm}$ to describe the enhanced carrier scattering due to the reduced size of the CF. In fact, based on the Fuchs–Sondheimer approximation [36], nanoscale resistivity can be estimated by

$$\rho = \rho_{\text{bulk}} \left(1 + \frac{3}{4} \frac{\lambda}{\phi} (1 - p) \right) \quad (7)$$

where λ is the electron mean free path, and p is the specularity factor, namely, the probability for elastic scattering at the CF surface. Using $\lambda = 28 \text{ nm}$ and $p = 0.5$ [36], one obtains $\rho = 380 \mu\Omega\text{cm}$ for Hf at $\phi = 1 \text{ nm}$, which is a realistic size of a small CF. The additional degradation of resistivity may be attributed to the phonon scattering due to the high temperature during electrical operation, surface roughness scattering, and scattering due to grain boundaries and other defects. Most notably, the CF composition in metal–oxide RRAM is not confirmed to consist of the fully reduced oxide (e.g., Hf in HfO_x); rather, suboxide phases have been evidenced based on temperature-dependent measurements [34] and electron microscopy [13], [14]. The resistivity of such suboxide phases may be far higher than their metallic counterparts [13].

A. Time Evolution of Set and Reset

Fig. 3 shows the calculated (a) R , (b) V , and (c) I as a function of time during set. The reset process is instead described in Fig. 3(d)–(f), showing the calculated R , V , and I , respectively, during reset. Fig. 3(e) and (f) shows the absolute values of voltage and current during reset, which are assumed to take place under negative voltage/current (see Fig. 1). In the simulations, the current limitation during set was provided by a 1T1R configuration [3], [4], [19]. According to a simple analytical model for the MOS transistor, the current in the saturation regime was given by $I_{\text{sat}} = K V_{\text{OD}}^2 / 2$, where $K = 700 \mu\text{Acm}^{-2}$, and V_{OD} is the overdrive voltage, namely, the gate voltage minus the threshold voltage of the MOS transistor. A voltage $V_A = 0.8 \text{ V}$ was applied to the 1T1R, resulting in

voltage drops V_{MOS} and V across the MOS transistor and the cell, respectively, with $V_A = V + V_{\text{MOS}}$. Three gate bias values of V_{OD} were used for the calculations in Fig. 3. The resistance decreases with time due to $R \propto \phi^{-2}$, where ϕ increases according to (6). The transient behavior of R in Fig. 3(a) features a fast growth regime starting around 1 μs , followed by a saturation regime. The onset of the saturation regime depends on I_C . The fast growth regime is due to the high local temperature after nucleation as a result of (5) and of the relatively large initial voltage V . The latter is virtually equal to V_A during the initial stages of the set transition due to initial high R of the CF. As R becomes comparable with the equivalent transistor resistance estimated by $R_{\text{MOS}} = V_A/I_C$, the voltage across the cell also decreases according to the voltage partition formula

$$V \approx V_A \frac{R}{R + R_{\text{MOS}}} \quad (8)$$

as evidenced in Fig. 2(b). As a result, T also decreases due to the quadratic relationship in (4), thus quenching the filament growth rate. The CF growth during set under a compliance system, consisting of a load resistance or transistor in series with the cell, can thus be viewed as a self-limiting process, where the increase of filament diameter results in an exponential decrease of the driving force for ion migration, thus suppressing growth [31]. Note that the current is practically constant and equal to $I_C = 7 \mu\text{A}$ for the smaller gate bias during the saturation regime, as shown in Fig. 3(c). The current corresponding to higher gate bias shows a marked change of slope at the transition between the fast R drop and saturation, without reaching a constant current because of the transistor not entering the saturation regime. In these cases, I_C was defined as the maximum current reached after the set time $t_{\text{set}} = 1 \text{ s}$. For increasing I_C , the saturated value of R decreases approximately as $R = V_C/I_C$, with $V_C \approx 0.4 \text{ V}$ representing the voltage across the cell at $t_{\text{set}} = 1 \text{ s}$.

The reset process was simulated assuming that the transistor was biased to a high V_{OD} to minimize the series resistance to a value $R_{\text{MOS}} = 10 \Omega$, thus allowing for a minimum voltage in the reset process. A quasi-static voltage sweep, where V_A linearly increases with time according to $V_A = -\beta t$, with $\beta = 2 \text{ Vs}^{-1}$, was assumed. The electric field reversal causes ions to migrate toward the opposite direction with respect to the set process, resulting in a decrease of the CF size. As a result, the resistance shown in Fig. 3(d) increases starting from a reset voltage V_{reset} and reset current $I_{\text{reset}} = V_{\text{reset}}/R$, which is apparent as the maximum of curves in Fig. 3(f). The reset voltage can be evaluated from the curves in Fig. 3(e) at the reset time, although no big change is seen due to V being almost equal to the applied voltage V_A given the negligible R_{MOS} during reset. As R increases, the current flowing through the filament strongly decreases. However, Joule heating in the model remains active since T is controlled by the voltage according to (4) and (5). Note that, more realistically, Joule heating may change during filament rupture due to the change of the thermal and electrical parameters k_{th} and ρ in (5) as a result of gap opening in the CF. In particular, the heating efficiency might degrade due to the thermal conductivity decreasing less than

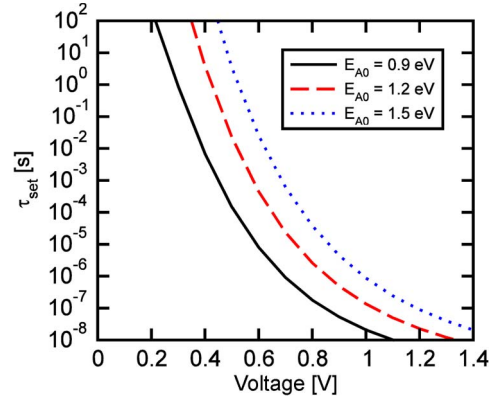


Fig. 4. (Color online) Calculated time for the onset of the set process τ_{set} as a function of voltage for increasing E_{A0} .

the electrical conductivity as a result of phonon heat conduction contribution in the low-conductive partially ruptured filament. This may result in a nonabrupt increase of R , as frequently observed in reset experiments on bipolar RRAM devices [2], [19], [21], [35]. The dissolution process in (6) was artificially stopped at a minimum diameter of 0.5 nm, which is equal to the initial diameter ϕ_N , to account for the presence of a residual low-conductivity filament after reset. Correspondingly, R tends to a finite value in Fig. 3(d). Note that the reset time (hence V_{reset}) slightly increases for decreasing R since large CFs require more time for complete dissolution [34].

The time for the onset of the set transition in Fig. 3(a)–(c) is about $\tau_{\text{set}} = 1 \mu\text{s}$ for an applied voltage $V_A = 0.8 \text{ V}$. This time marks the onset of the change of diameter of the CF and is due to the ion migration kinetics in (6). This equation allows deriving an analytical formula for τ_{set} given by

$$\tau_{\text{set}} = A \Delta \phi e^{\frac{E_{A0} - \alpha q V}{k T_0 \left(1 + \frac{V^2}{8 T_0 \rho k_{\text{th}}} \right)}} \quad (9)$$

where $\Delta \phi$ is the change of CF diameter required for a sufficient change of resistance. Fig. 4 shows the calculated τ_{set} as a function of the cell voltage V according to (9) using $\Delta \phi = 0.5 \text{ nm}$, thus equal to the initial CF size, and parameter values equal to those used in Fig. 3, except for a variable activation energy $E_{A0} = 0.9, 1.2$ and 1.5 eV . The cell voltage was assumed equal to the applied voltage V_A , which, according to (8), is acceptable in the initial stages of the set process when R is significantly larger than R_{MOS} . For $V_A = 0.8 \text{ V}$ and $E_A = 1.2 \text{ eV}$, one gets $\tau_{\text{set}} = 2.6 \mu\text{s}$, in agreement with Fig. 3(a)–(c). Note the significant dependence on V due to the temperature-activated growth process and on the activation energy, which also enters the exponent in (6): τ_{set} exponentially increases with E_{A0} due to the enhanced energy barrier for ion migration. It is useful to recall that the nucleation time for the formation of the initial CF embryo in Fig. 2(b) is neglected in (1), (6), and (9). In cases of set transitions, where an initial CF is not present, nucleation might require an additional time; thus, τ_{set} might be underestimated in Fig. 4. On the other hand, neglecting nucleation does not significantly affect the calculated R at the end of the set transient [Fig. 3(a)] and the corresponding reset current [Fig. 3(f)]. In fact, at sufficiently high V_A , the nucleation time is so short that it can be neglected with respect to the

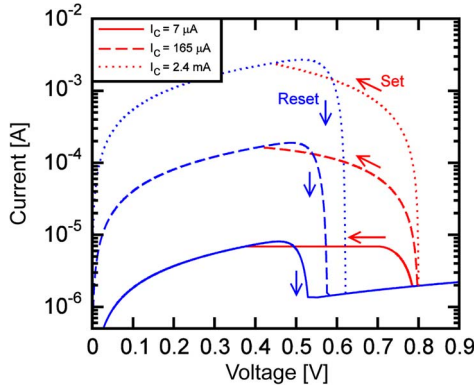


Fig. 5. (Color online) Calculated I - V curves for set and reset at increasing I_C obtained from the calculated V and I in Fig. 3.

overall set time t_{set} , generally in the saturation regime in Fig. 3. Therefore, neglecting nucleation, although possibly leading to an underestimated τ_{set} in Figs. 3 and 4, will not affect the conclusions about the set state R , the reset current, and their universal behavior in this paper.

B. Current-Voltage Characteristics

Fig. 5 shows the I - V curves during set and reset transitions, which were obtained from the calculated V and I in Fig. 3. The set transition is controlled by the saturation level of the MOSFET, with the voltage decreasing to about $V_C = 0.4$ V at the end of set at $t_{\text{set}} = 1$ s, as already seen in the saturation regime of Fig. 3(b). This is the result of the self-limiting process of set with a current compliance, resulting in a voltage regulation at any time during set. During reset, the current increases along the linear characteristic of the device until the filament dissolution takes place at an approximately constant V_{reset} around 0.5 V, slightly larger than V_C . The reset current is slightly higher than I_C , with a ratio $\eta = I_{\text{reset}}/I_C \approx 1.1$; this is due to the comparable timescales of set and reset operations, both around 1 s in these calculations. The slightly faster reset and the different pulse shape (triangular reset pulse versus rectangular set pulse) result in I_{reset} being larger than I_C due to the time-voltage tradeoff in (1), where a shorter time requires a higher temperature, hence voltage, for reset [37].

Fig. 6 shows the resistance R after (a) the set process and (b) the reset current I_{reset} as a function of I_C . The figure compares data from various literature sources for oxide-based RRAM showing unipolar (NiO [4], [20]) and bipolar switching (HfO_x [19], TiO_x [21] and ZrO_x/HfO_x [2]) compared with calculations from the ion-migration model. The calculated R in Fig. 6(a) was taken at $t_{\text{set}} = 1$ s, corresponding to the typical time in quasi-static measurements considered in the figure. The physical parameters were the same as for the calculations in Fig. 3, except for a variable activation energy $E_{A0} = 0.9, 1.2$, and 1.5 eV. Both calculated R and I_{reset} display the same I_C dependence as in the experimental behavior, namely, $R = V_C I_C^{-1}$ with $V_C \approx 0.4$ V, and $I_{\text{reset}} = \eta I_C$ with $\eta \approx 1.1$. The latter relationship is due to $I_{\text{reset}} = V_{\text{reset}}/R = V_{\text{reset}}/V_C \times I_C$, with $V_{\text{reset}}/V_C = \eta \approx 1.1$ due to the slightly different timescales and pulse shapes during set and reset operations. The almost equal I_{reset} and I_C are therefore a consequence of

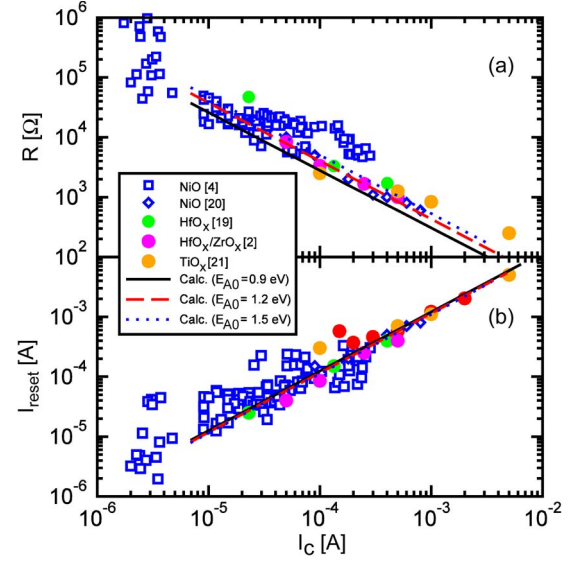


Fig. 6. (Color online) Measured and calculated (a) R and (b) I_{reset} as a function of I_C . Data were obtained from unipolar [4], [20] and bipolar RRAMs [2], [19], [21] under limited current during set. Calculations were performed for an applied voltage $V_A = 0.8$ V and variable activation energies $E_{A0} = 0.9, 1.2$, and 1.5 eV. The weak dependence on E_{A0} is the result of the voltage regulation in the negative feedback loop of (6) and accounts for the observed universal set/reset characteristics in RRAM [25].

the similarity of values of V_{reset} and V_C , which are due to (1) dictating both set and reset transitions with the same values of E_{A0} , α , and A .

The independence of set/reset characteristics from E_{A0} in Fig. 6 is consistent with the experimental finding of a nearly constant V_C and η for most metal oxides investigated for RRAM applications [25]. However, it apparently conflicts with the marked E_{A0} dependence of τ_{set} in Fig. 4. The E_{A0} -independent universal nature of the set process can be understood by the self-limiting CF growth process, where the voltage is controlled by R [(8)], which in turn changes with time at a V -dependent rate [(6)]. In fact, the filament growth results in a decrease of R , causing a decrease of V in (8), which exponentially suppresses the CF growth rate according to (6). This results in a negative feedback loop that regulates V across the cell at any time during the set, with small dependences on the physical parameters of the CF (e.g., thermal and electrical conductivities) and of the set/reset dynamics, as evidenced by the weak dependence on E_{A0} , in Fig. 6. Similarly, weak changes were found for the calculated set/reset characteristics as a function of A and α in (6).

Note that E_{A0} , A , and α are microscopic local parameters that are subjected to statistical variations from cell to cell in the array or in the wafer and from cycle to cycle due to the stochastic nature of ion migration resulting in different shapes and structures of the CF during set/reset cycling. Thus, the small changes of set/reset kinetics in response to changes of microscopic parameters (e.g., E_{A0} in Fig. 6) might explain the observed statistical spread of switching parameters, such as set/reset voltage and set/reset state resistance [2], [19].

V. OVERSHOOT EFFECTS

It has been reported that imperfect limitations of I_C due to a high parasitic capacitance C_p of the RRAM cell or to

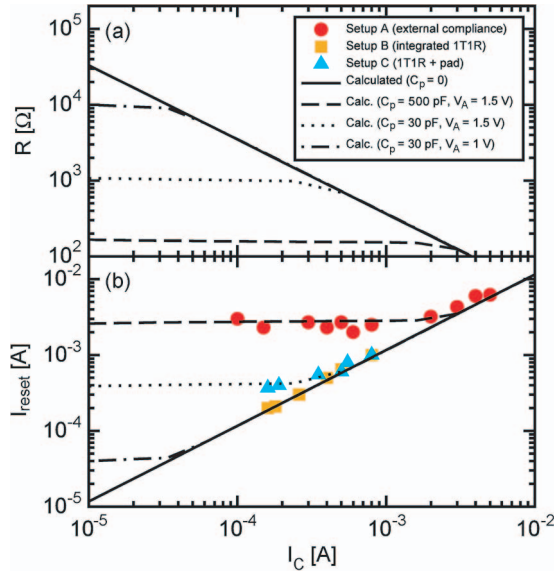


Fig. 7. (Color online) Measured and calculated (a) R and (b) I_{reset} as a function of I_C , describing overshoot effects due to parasitic capacitance C_p or relatively slow compliance systems. Data were obtained from unipolar switching with different configurations, namely, external compliance in (A) semiconductor parameter analyzer, (B) 1T1R, or (C) 1T1R with an additional pad at the cell TE [3]. Calculations were performed assuming $C_p = 30$ or 500 pF and $V_A = 1$ or 1.5 V. Saturation effects due to the current overshoot become more important for increasing C_p and V_A .

an insufficiently fast response of the current compliance in quasi-static measurements can result in poor control of R and I_{reset} [3], [30], [38]. To clarify the impact of the parasitic capacitance and compliance adjustment time on the set transition, the set/reset properties of a cell with different C_p were calculated from the model of (6). Fig. 7 shows the calculated (a) R and (b) I_{reset} as a function of I_C , assuming the same physical parameters as in Fig. 6 with $E_{A0} = 1.2$ eV, and for different values of the applied voltage V_A and of C_p . Data for I_{reset} are also shown corresponding to different setup configurations: namely, (A) an external compliance where the voltage was automatically reduced at switching to maintain a constant I_C ; (B) an integrated 1T1R structure, where I_C was directly controlled by the transistor in series with the RRAM cell, and the transistor and cell were integrated in the same chip to minimize the parasitic capacitance of the cell; and (C) a 1T1R structure modified by the presence of an electrical contact pad to allow probing the potential drop across the cell during the measurement [3]. Note that B and C coincide except for an enhanced C_p in configuration C due to the pad capacitance. The reset current saturates at about 2–3 mA for configuration A at I_C smaller than 2 mA, while no saturation is apparent for setup B. Configuration C shows evidence for saturation starting at about $I_{\text{reset}} \approx I_C \approx 300$ μ A. Calculations predict a resistance saturation to R_{sat} and, consequently, a saturation of I_{reset} due to $I_{\text{reset}} = V_{\text{reset}}/R$ with an almost constant V_{reset} . The experimental behavior of configuration A is reproduced by calculations for $V_A = 1.5$ V, corresponding to the experimental set voltage V_{set} in [3], and $C_p = 500$ pF. The latter can be viewed as an effective parasitic capacitance describing the time delay required by the external compliance to adjust the voltage soon after the onset of the set transition. The effective delay time

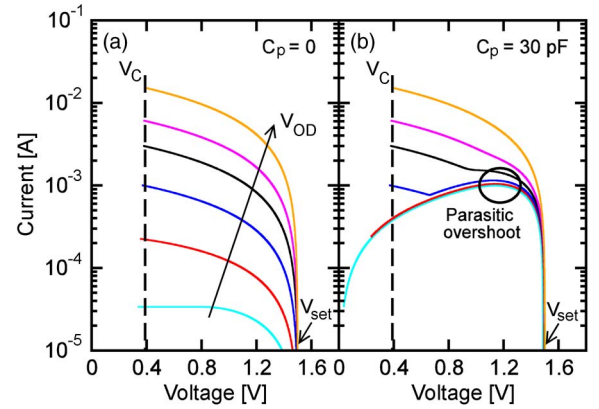


Fig. 8. (Color online) Calculated I - V curves for (a) $C_p = 0$ and (b) $C_p = 30$ pF at increasing gate bias V_{OD} of the 1T1R. The overshoot effects and its consequence on the final value of R are apparent from (b) at low gate bias.

can be estimated from $\tau_{\text{RC}} = R_{\text{MOS}}C_p = V_A C_p / I_C$. Using the value $I_C = 2$ mA for the onset of the I_{reset} saturation, one obtains a delay time of around 0.4 μ s, in reasonable agreement with the measured delay in a standard semiconductor parameter analyzer HP 4155 [39]. Data for setup configuration C can be explained assuming a smaller capacitance $C_p = 30$ pF, yielding a delay time at saturation of around 150 ns. If a smaller voltage is applied, thanks to a smaller V_{set} of the RRAM sample, then R and I_{reset} saturate at a smaller I_C , as shown by calculations in Fig. 7 for $C_p = 30$ pF and $V_A = 1$ or 1.5 V. This is consistent with experiments of current-limited forming as a function of the forming voltage, where lower overshoot effects were detected for relatively low forming voltages [38].

To understand the C_p -dependent saturation of R and I_{reset} in Fig. 7, Fig. 8 shows the calculated I - V curves during set for (a) $C_p = 0$ and (b) $C_p = 30$ pF for increasing the gate bias of the MOS transistor. The I - V curves in Fig. 8(a) show the usual voltage-driven transition from the set voltage V_{set} to $V_C = 0.4$ V after 1 s from the onset of the set transition. Fig. 8(b) shows identical transient I - V curves only for relatively high I_C above 5 mA, whereas a clear current overshoot is shown for smaller I_C values. The overshoot is due to R quickly decreasing due to the CF growth, thus resulting in an increase of current at an almost constant voltage, due to the delayed discharge of the parasitic capacitance in parallel to the cell [30], [40]. The delayed voltage response breaks the negative feedback loop and results in an overgrowth of the CF for short set times. As R decreases, V decreases due to the accelerating discharge of C_p , thus soon suppressing any further CF growth. The value of the resistance after this first set transient is the saturated resistance R_{sat} in Fig. 7 and depends on the CF growth kinetics, on C_p , and on the applied voltage. Kinetic parameters inducing fast growth (e.g., low E_A , high A), a large C_p , and a large V_A enhance the tendency to overshoot and the consequent saturation of R and I_{reset} . As the CF growth is interrupted by the voltage decrease, the current decreases along the linear characteristic $I = V/R_{\text{sat}}$ according to the approximated RC discharge characteristic

$$I(t) \approx \frac{V_A}{R_{\text{sat}}} e^{-\frac{t}{R_{\text{sat}} C_p}}. \quad (10)$$

According to (10), the current would reach I_C at $t_C = R_{\text{sat}} C_p \log(V_A I_C^{-1} R_{\text{sat}}^{-1})$. If $t_C < t_{\text{set}}$ [which is the case for the two intermediate curves in Fig. 8(b)], then the usual voltage-controlled set dynamics can be established again, and no saturation effects will eventually be detected. Otherwise, R remains stuck at R_{sat} with the voltage reaching a relatively low value given by $R_{\text{sat}} I_C \ll V_C$ at t_{set} . This analysis points out that overshoot effects also depend on the total set pulse width, with relatively long set operations being less prone to saturation.

VI. DISCUSSION

The proportionality between R and I_C in the set characteristic of Fig. 6 is due to the constant voltage V_C reached at the end of the set operation, which is a consequence of the voltage regulation effect of the feedback loop in (6) and (8). Note that (6) contains a critical assumption, namely, $R \propto R_{\text{th}} \propto \phi^{-2}$, and this is responsible for the achievement of an inverse proportionality between R and I_C . Other models for R and R_{th} would have not been comparably able to account for the experimental results. For instance, assuming a constant R_{th} regardless of the filament size would have resulted in a constant final power $P_C = V^2/R = RI^2$, resulting in a slope of -2 in the $R - I_C$ plot in Fig. 6. These results suggest therefore that the thermal model with $R \propto R_{\text{th}}$ is most appropriate for the CF, and that this is the key to explain the observed constant RI_C in set experiments. We also note that $R \propto R_{\text{th}}$ was also derived for a large resistance range in CFs in NiO based on the study of V_{reset} [34]. The proportionality between R and R_{th} indicates that electrical and thermal conductions are both dominated by carrier transport along the filament, with small contributions from phonons and/or out-of-filament heat loss. This should at least be the case for relatively low local temperatures, corresponding to the final stages of the set transition, when the voltage becomes comparable with V_C .

The bipolar switching model in this paper might be extended to describe set/reset in conductive bridge RAM (CBRAM) devices. These can be viewed as bipolar RRAM devices where one inert metal in one electrode is replaced by an active metal, usually consisting of fast-migrating Cu or Ag [1]. Analytical models based on thermally activated filament growth were proposed for CBRAM [31], [32], similar to the bipolar metal-oxide RRAM model considered here. However, a different thermal model was used in these works, where a fixed thermal resistance $R_{\text{th}} = 10^5 \text{ KW}^{-1}$ was assumed as a fitting parameter. This resulted in a relatively small power-driven heating of a few hundred degrees in the CF [31]. Voltage-driven set/reset behavior was obtained in these models by the energy barrier lowering term similar to (2), thus resulting in constant-voltage set [31] and reset [32] characteristics similar to Fig. 6.

The filament model that has been adopted in this paper relies on the concept of a metallic filament, where the electric and thermal resistances are dictated solely by the size ϕ [see (5)]. This model is simplified since a metallic conductivity is known to apply only at relatively small resistances $R < 1 \text{ k}\Omega$, whereas a semiconductor behavior of resistance, where conduction takes place by thermally activated hopping through defects, is generally observed for higher R [7]. Due to this simplification, we should view the calculated ϕ in this paper only as an effective

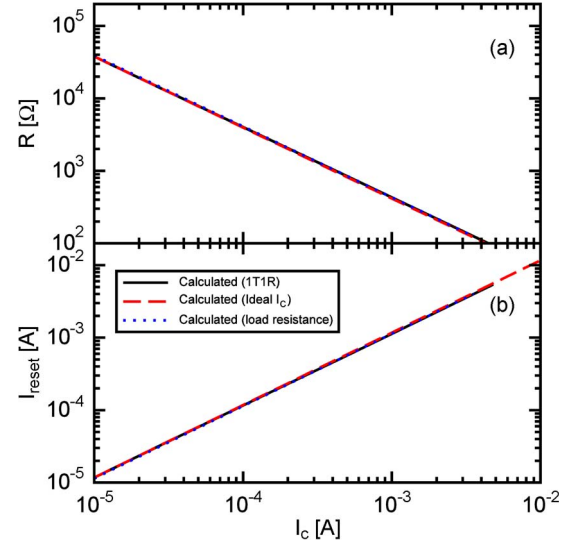


Fig. 9. (Color online) Calculated (a) R and (b) I_{reset} as a function of I_C depending on the setup configuration during set, namely, a 1T1R structure, an ideally constant I_C , and a load resistance. No parasitic capacitance was assumed in the calculations.

filament size, namely, the value of ϕ that would yield the measured resistance assuming a metallic conductivity ρ of the filament. This definition might result in unphysical values of the effective filament size, particularly for high resistance. It is important to note that this assumption is not necessary to obtain the key result in this paper, namely, a constant $V_0 = RI_C$ at the end of the set process. In fact, a different filament model, e.g., assuming a semiconductor transport with a positive activation energy E_{AR} [7], would result in the same relationship between resistance at the end of the set process and I_C . In fact, the filament growth model in (1) can be generalized into a rate of R decrease given by

$$\frac{dR}{dt} = -A_R e^{-\frac{E_{\text{AR}}}{k(T_0 + R_{\text{th}} R I^2)}} \quad (11)$$

where E_{AR} and A_R are the activation energy and the preexponential factor for resistance change. These may be different from E_A and A in (1) since the rate of change of resistance may include not only the geometrical growth of ϕ but also the increase of ion/defect concentration acting as a dopant within the CF, thus changing the Fermi level and the corresponding activation energy for conduction [7], [34], [41], [42]. Assuming a proportionality between electrical and thermal resistances with $R_{\text{th}} = R(8k_{\text{th}}\rho)^{-1}$, one obtains a similar voltage regulation effect as in (6), thus leading to a constant V_C at the end of the set transition.

It is useful to discuss the results of this model with reference to alternative experimental setup and/or programming schemes. For instance, the experimental results in Fig. 6 have been obtained not only with 1T1R structures [3], [4], [19], [20] but also using a compliance current approach, where the measurement equipment was able to readjust the voltage across the device to maintain a given maximum current I_C [3], [22]–[24]. The model predicts very similar evolutions of the resistance with time, irrespective of how the current is limited during the set transition. To demonstrate this point, Fig. 9 shows the

calculated (a) R and (b) I_{reset} as a function of I_C for different setup schemes, namely, a MOS transistor (same results as in Fig. 6), an ideal compliance system keeping a constant current I_C above the set point, and a variable load resistance. For non-constant current compliance, I_C was defined as the maximum current at the end of set at $t_{\text{set}} = 1$ s. The same applied voltage $V_A = 0.8$ V was used in the calculations. Simulation results show that R and I_{reset} are negligibly affected by the different choice of the setup, thus justifying the experimental observation of the same universal set/reset characteristics independent of the setup configuration [25].

Finally, although this model should mainly be applied to bipolar RRAM, where the reversible set/reset mechanisms of filament growth and narrowing occurs, calculations have been shown to account also for unipolar RRAM data in Fig. 6. This suggests that unipolar switching may be described by a CF growth process similar to the bipolar switching model in Fig. 2. However, the fact that set and reset are both achieved under the same polarity in unipolar switching rules out ion migration as the leading process for both set and reset. Assuming that ion migration has a major role in CF narrowing during reset, one should thus provide physical explanations for CF growth under unipolar RRAM. A possible explanation is the chemical driving force based on metal–oxide reduction, which takes place at high temperatures because of the free energy for oxide formation being positive at sufficiently high temperature [43], [44]. The oxide reduction reaction and the consequent metallic phase segregation are assisted by atomic/ionic migrations, which might occur by either 1) thermally activated electromigration, due to the electrons flowing from the bottom to the TE, thus pushing ions in the direction opposite to the one indicated in Fig. 2, or 2) thermally activated diffusion, where the strong vertical concentration gradient of conductive atoms drives diffusion until the CF is reconnected by atomic flow of atoms in the direction opposite to Fig. 2. The thermally activated nature of the set phenomena results in a growth equation similar to (1), thus similarly accounting for the voltage-driven set dynamics at the basis of the constant $V_C = RI_C$ for unipolar RRAM in Fig. 6 [44].

VII. CONCLUSION

An ion migration model for filament growth/dissolution has been shown to account for the universal set/reset characteristics in bipolar RRAM devices based on binary metal oxides. The model accounts for the constant voltage at the end of the set transition due to the voltage regulation in a negative feedback loop in the presence of a current limiting scheme. The numerical model based on this physical mechanism is used to study the dependence on activation energy and configuration setup, evidencing the different impact of current overshoot depending on the parasitic capacitance and the applied voltage during set.

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